

Chapter 8 Heteroskedasticity

- 5 The variable *smokes* is a binary variable equal to one if a person smokes, and zero otherwise. Using the data in SMOKE.RAW, we estimate a linear probability model for *smokes*:

$$\begin{aligned} \widehat{smokes} = & .656 - .069 \log(cigpric) + .012 \log(income) - .029 educ \\ & (.855) \quad (.204) \quad (.026) \quad (.006) \\ & [.856] \quad [.207] \quad [.026] \quad [.006] \\ & + .020 age - .00026 age^2 - .101 restaurn - .026 white \\ & (.006) \quad (.00006) \quad (.039) \quad (.052) \\ & [.005] \quad [.00006] \quad [.038] \quad [.050] \\ n = 807, R^2 = .062. \end{aligned}$$

The variable *white* equals one if the respondent is white, and zero otherwise; the other independent variables are defined in Example 8.7. Both the usual and heteroskedasticity-robust standard errors are reported.

- (i) Are there any important differences between the two sets of standard errors?
- (ii) Holding other factors fixed, if education increases by four years, what happens to the estimated probability of smoking?
- (iii) At what point does another year of age reduce the probability of smoking?
- (iv) Interpret the coefficient on the binary variable *restaurn* (a dummy variable equal to one if the person lives in a state with restaurant smoking restrictions).
- (v) Person number 206 in the data set has the following characteristics: *cigpric* = 67.44, *income* = 6,500, *educ* = 16, *age* = 77, *restaurn* = 0, *white* = 0, and *smokes* = 0. Compute the predicted probability of smoking for this person and comment on the result.

Suggested Answer

- (i) No. For each coefficient, the usual standard errors and the heteroskedasticity-robust ones are practically very similar.
- (ii) The effect is $-.029(4) = -.116$, so the probability of smoking falls by about .116.
- (iii) As usual, we compute the turning point in the quadratic: $.020/[2(.00026)] \approx 38.46$, so about 38 and one-half years.
- (iv) Holding other factors in the equation fixed, a person in a state with restaurant smoking restrictions has a .101 lower chance of smoking. This is similar to the effect of having four more years of education.
- (v) We just plug the values of the independent variables into the OLS regression line:

$$smôkes = .656 - .069 \cdot \log(67.44) + .012 \cdot \log(6,500) - .029(16) + .020(77) - .00026(77^2) \approx .0052.$$

Thus, the estimated probability of smoking for this person is close to zero. (In fact, this person is not a smoker, so the equation predicts well for this particular observation.)